

Ramadhan Adam Zome

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RESEARCH

Open-world Windows malware detection using self-supervised byte-level embeddings and prototypical few-shot adaptation. Broader interests: representation learning, AI security, low-resource NLP, and philosophy of mind.

EDUCATION

MSc Artificial Intelligence and Machine Learning

2024 – present

PAUSTI (Pan African University, hosted at JKUAT), Nairobi

Supervisor: Dr. Albert Njoroge Kahira · Reg: MA300-0003/2024

BSc Statistics and Programming — First Class

2020 – 2024

Kenyatta University, Nairobi

PUBLICATIONS & PREPRINTS

Geometric Sparse Attention via Radial Basis Function Graph Traversal

2026

Ramadhan Adam Zome. *arXiv preprint, cs.LG*, 2026.

Proposes RBF kernel as a replacement for scaled dot-product attention. Novel contributions: gradient analysis showing implicit metric learning, Swahili low-resource benchmark, honest FFN redundancy negative result.

SELECTED PROJECTS

Deepfake Detection + RAG Explainability System

2025 – 2026

- Xception CNN trained from scratch on 140k real/fake faces (Kaggle dataset).
- RAG pipeline built from scratch (no LangChain): FAISS + sentence-transformers + Groq llama-3.3-70b.
- FastAPI backend on HuggingFace Spaces (Docker); React/Vite frontend on Vercel.

Swahili GPT

2025

- Character-level GPT trained from scratch on 65M characters of Swahili news (29,544 articles).
- Published as Medium article; GitHub: [RamadhanAdam/swahili-gpt](https://github.com/RamadhanAdam/swahili-gpt) with CITATION.cff.

BirdSpeak — Acoustic Token Transformer

2025

- Transformer encoder on MFCC + k-means quantized acoustic tokens for bird call classification.
- Joint masked acoustic modelling + classification objectives. Ablation vs MLP and LSTM baselines.

AI From First Principles — C Implementations

2024 – present

- Neural network implementations in C: Elman RNN trained with a genetic algorithm + raylib visualizer.
- Public Medium series covering foundational models from scratch; emphasis on internals over frameworks.

WORK EXPERIENCE

Data Analyst & Cybersecurity Associate

Dec 2023 – Jul 2025

Innovative Toll Solutions, Nairobi

- Data cleaning, EDA, and modelling on operational logs and transaction datasets.
- Built dashboards and reporting pipelines for operations teams.
- Log-level anomaly detection, alert triage, and traffic pattern identification.
- Collaborated with engineers on ingestion pipeline resilience after DDoS incidents.

TECHNICAL SKILLS

Languages: Python, C, SQL, Bash

ML / DL: PyTorch, TensorFlow, Scikit-learn, HuggingFace Transformers, FAISS

NLP: Tokenisation, n-gram LMs, Viterbi POS tagger, dependency parsing, BERT fine-tuning

Security: Malware analysis, reverse engineering, PE static feature analysis, sandboxing (REMnux, FLAREVM)

Tools: Git, Linux, Docker, FastAPI, React/Vite, Jupyter, Tableau

RELEVANT COURSEWORK

Robotics (IK, Lagrange-Euler dynamics, D-H parameters) · Computer Vision & Pattern Recognition (PCA, LDA, Harris, ORB, HOG) · NLP (Arc-Eager dependency parser, LAS >0.81 on EN/DA/SV) · Deep Learning (BERT fine-tuning, miniature BERT encoder from scratch, LSTM text generation) · Ethics in AI

ACTIVITIES & SERVICE

- Applied to Deep Learning Indaba 2026 (Lagos, August 2026).
- Writing AI From First Principles on Medium (@ramadhanzome4) — public technical education series.
- Independent reading program: 20 landmark AI papers chronologically (McCulloch & Pitts 1943 → present).
- AWS AI & ML Scholars program participant.

References available upon request.